



WORK SHEET - 30

QUADRATIC EQUATION & LOGARITHM

Q.1 (D)

$$2\log_2 x^2 = \log_{x^2} 2^y \Rightarrow 4\log_2 x = \frac{y}{2}\log_x 2 \Rightarrow 8 = y(\log_x 2)^2 \Rightarrow 8 = y^3$$

$$\Rightarrow y = 2 \Rightarrow 2 = \log_x 2 \Rightarrow x = \sqrt{2}$$

$$\therefore x_0 = \sqrt{2} \text{ and } y = 2. \text{ Ans.}$$

Q.2 (B)

Let equation is $K(x - \alpha)(x - \beta) = 0$ where α, β are prime numbers
Sum of coefficients $= K - K(\alpha + \beta) + K\alpha\beta = K(\alpha - 1)(\beta - 1)$ prime
 $\Rightarrow K = 1$ and one of $(\alpha - 1)$ and $(\beta - 1)$ will be one
Let $\alpha - 1 = 1 \Rightarrow \alpha = 2$, $\beta - 1$ will be 2 So $\beta = 3$
Hence sum of roots 5. **Ans.**

Q.3 (C)

We have

$$S_1 = \sum x_1 = \sin 2\beta, \quad S_2 = \sum x_1 x_2 = \cos 2\beta$$

$$S_3 = \sum x_1 x_2 x_3 = \cos \beta, \quad S_4 = x_1 x_2 x_3 = -\sin \beta$$

$$\text{So that } \sum_{i=1}^4 \tan^{-1} x_i = \tan^{-1} \frac{S_1 - S_3}{1 - S_2 + S_4} = \tan^{-1} \frac{\sin 2\beta - \cos \beta}{1 - \cos 2\beta - \sin \beta} = \tan^{-1} \frac{\cos \beta (2 \sin \beta - 1)}{\sin \beta (2 \sin \beta - 1)}$$

$$= \tan^{-1}(\cot \beta) = \tan^{-1} \left(\tan \left(\frac{\pi}{2} - \beta \right) \right) = \frac{\pi}{2} - \beta. \text{ Ans.}$$

Q.4 (C)

Let $x^4 + a_1 x^3 + a_2 x^2 + a_3 x + 5 = 0$ 

$$\sum \alpha = -a_1; \sum \alpha\beta = a_2; \sum \alpha\beta\gamma = -a_3 \text{ and } \alpha\beta\gamma\delta = 5$$

Using A.M. $\geq$ G.M.

$$\left(\frac{\alpha + \beta + \gamma + \delta}{4} \right) \left(\frac{\sum \alpha\beta\gamma}{4} \right) \geq (\alpha\beta\gamma\delta)^{\frac{1}{4}} (\alpha^3 \beta^3 \gamma^3 \delta^3)^{\frac{1}{4}}$$

$$\left(\frac{-a_1}{4} \right) \left(\frac{-a_3}{4} \right) \geq (\alpha\beta\gamma\delta)$$

$$\frac{a_1 a_3}{16} \geq 5$$

$$a_1 a_3 \geq 80$$



Q.5 (C)
 $f(x) = -3$
 $x^2 - 2x - 3 = 0 \Rightarrow x = 0, 2$. **Ans.**

Q.6 (A)
 Root's of the equation $x^2 + 3x + 5 = 0$ are imaginary
 So both root's common $\Rightarrow a = 9$; $c = 15$
 a, b, c are in AP $\Rightarrow b = 12$
 Inradius of ΔABC is $r = \frac{\Delta}{s} = 3$. **Ans.**

Q.7 (A)
 Let $y = f(x) = \frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} = \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x+4)}$
 $y = \frac{x-3}{x+4}$ ($x \neq 1$; $x \neq 2$)

$xy + 4y = x - 3$
 $x(y-1) = -4y-3$
 $x = \frac{4y+3}{1-y}$; $y \neq 1$; $y \neq f(1)$; $y \neq f(2)$
 so no solution for 3 values.

Q.8 (B)
 Let $f(x) = x^2 - (a+1)x + (a^2 + a - 8)$



So, $f(2) < 0 \Rightarrow -2 < a < 3$ **Ans.**

Q.9 (C)
 Sum of roots $= \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3 + \beta^3}{\alpha\beta} = \left(\frac{-p}{q} \right)$

Product of roots $= \frac{\alpha^2}{\beta} \cdot \frac{\beta^2}{\alpha} = \alpha\beta = q$;

So, $x^2 - (\text{sum of root})x + (\text{Product of roots}) = 0$
 $\Rightarrow qx^2 + px + q^2 = 0$ **Ans.**

Q.10 (D)
Case-I : $x^2 > 1$, then $\frac{x}{|x|} - x \geq 1 \Rightarrow x \in (-\infty, -2]$

Case-II : $0 < x^2 < 1$, then $0 < \frac{x}{|x|} - x \leq 1 \Rightarrow x \in (0, 1)$

$\therefore$ We get, $x \in (-\infty, -2] \cup (0, 1)$ **Ans.**

Q.11 (B)

$$x^2 + 2x - n = 0 \Rightarrow (x+1)^2 = n+1 \Rightarrow x = -1 \pm \sqrt{n+1}$$

Thus, $(n+1)$ should be a perfect square.

$$\therefore n = 8, 15, 24, 35, 48, 63, 80, 99$$

$\Rightarrow$ number of values of n equals 8.

Q.12 (C)

$$\cos^4 \frac{\pi}{8} + a \cos^2 \frac{\pi}{8} + b = 0$$

$$\text{As, } \cos \frac{\pi}{4} = \left(2 \cos^2 \frac{\pi}{8} - 1 \right) \Rightarrow \cos^2 \frac{\pi}{8} = \left(\frac{1}{\sqrt{2}} + 1 \right) \frac{1}{2}$$

$$\text{So, } \cos^4 \frac{\pi}{8} = \frac{1}{4} \left(\frac{1}{2} + 1 + \frac{2}{\sqrt{2}} \right) = \left(\frac{3}{2} + \sqrt{2} \right) \frac{1}{4}$$

$$\text{Thus, } \left(\frac{3}{8} + \frac{a}{2} + b \right) + \sqrt{2} \left(\frac{1}{4} + \frac{a}{4} \right) = 0$$

$$\text{As, } a, b \in \mathbb{Q}, \text{ so } \frac{1}{4} + \frac{a}{4} = 0, \frac{3}{8} + \frac{a}{2} + b = 0 \Rightarrow a = -1, b = \frac{1}{8} \quad \text{Ans.}$$

Q.13 (A)

$$x^2 + b^2 + 2bx = 1 \Rightarrow x = \pm 1 - b$$

$$x^2 + a^2 + 2ax = 1 \Rightarrow x = \pm 1 - a$$

$$\Rightarrow 1 - b = 1 - a \Rightarrow a = b$$

$$\text{or } 1 - b = -1 - a \Rightarrow -a = 2 - b$$

$$\text{or } -1 - b = 1 - a \Rightarrow a = 2 + b$$

$$\text{or } -1 - b = -1 - a \Rightarrow a = b$$

Clearly, the possibility of $a = b$ is ruled out. Because in this case, equations will have two common roots.

Thus, $a = b + 2$.

Q.14 (B)

$$\text{Let } t = x^2 + 2xy + 3y^2 - 6x - 2y$$

$$\Rightarrow x^2 + 2(y-3)x + (3y^2 - 2y - t) = 0$$

$$\text{As, } x \in \mathbb{R} \Rightarrow D \geq 0$$

$$\Rightarrow 4(y-3)^2 - 4(3y^2 - 2y - t) \geq 0$$

$$\Rightarrow -2y^2 - 4y + 9 + t \geq 0$$

$$\Rightarrow t \geq 2(y+1)^2 - 11 \Rightarrow 2y^2 + 4y - (9+t) \leq 0$$

$$\text{put } D \geq 0$$

$$\Rightarrow t \geq -11$$

Thus, the minimum value of given expression is -11 .

Q.15 (A)

$$\Sigma \alpha = 1, \quad \Sigma \alpha\beta = 0, \quad \alpha\beta\gamma = 1$$

$$\text{Now, } \Sigma \frac{1+\alpha}{1-\alpha} = \Sigma \frac{2-(1-\alpha)}{1-\alpha} = \Sigma \left(\frac{2}{1-\alpha} - 1 \right) = 2 \Sigma \frac{1}{1-\alpha} - 3$$

$$\text{Now, } \frac{1}{x-\alpha} + \frac{1}{x-\beta} + \frac{1}{x-\gamma} = \frac{(3x^2 - 2x)}{x^3 - x^2 - 1}$$

Put $x = 1$, we get $\frac{1}{1-\alpha} + \frac{1}{1-\beta} + \frac{1}{1-\gamma} = -1$

$\therefore$ The given expression equals -5 **Ans.**

Q.16 (B)

Q.17 (A)

(i) $ax^2 + a^2x + 2 = 0$ has imaginary or equal roots if

$$(a^2)^2 - 4(2)(a) \leq 0$$

i.e., $a(a^3 - 8) \leq 0$

i.e., $a(a-2)(a^2 + 4 + 2a) \leq 0$

i.e., $a \in (0, 2]$ (as $a \neq 0$)

$\therefore S \equiv (0, 2]$

Number of integral values in set S is 2

(ii) $-\alpha\beta = -\left(\frac{-a^2}{a}\right)\left(\frac{2}{a}\right) = 2$

Prime number $\forall a \in \mathbb{R}$. **Ans.**

Q.18 (D)

Q.19 (B)

$$f(n) = \sum_{r=C_2}^n \frac{r}{rC_2} = \sum_{r=2}^n \left(\frac{1}{rC_2} - \frac{1}{r+1C_2} \right) \quad (\text{As } {}^{r+1}C_2 - {}^rC_2 = r)$$

$$f(n) = 1 - \frac{1}{n+1C_2} \dots \dots \dots [\text{As } {}^nC_1 + {}^nC_2 = {}^{n+1}C_2]$$

$$f(n) = 1 - \frac{2}{n(n+1)} \dots \dots \dots \quad a = \lim_{n \rightarrow \infty} f(n) = 1$$

(i) $f(7) = 1 - \frac{2}{7 \times 8}$ and also $f(8) = 1 - \frac{2}{8 \times 9}$

$\therefore f(7) + f(8) = \frac{122}{63}$

(ii) $x^2 - \frac{3x}{2} + t = 0 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right\rangle ; \quad \alpha + \beta = \frac{3}{2}$

A.M $\geq$ H.M

$$\frac{\frac{\alpha}{2} + \frac{\alpha}{2} + \beta}{3} \geq \frac{3}{\frac{2}{\alpha} + \frac{2}{\alpha} + \frac{1}{\beta}} \Rightarrow \frac{4}{\alpha} + \frac{1}{\beta} \geq 6.$$

Q.20 (AD)

We have $x_1 + x_2 = 3$; $x_1x_2 = \alpha$

$x_3 + x_4 = 12$; $x_3x_4 = \beta$

Now, let $x_1 = a$; $x_2 = ar$; $x_3 = ar^2$; $x_4 = ar^3$

$\therefore$ The above equation becomes $a + ar = 3$; $a(ar) = \alpha$

$$ar^2 + ar^3 = 12, ar^2 \cdot ar^3 = \beta$$



Now, we get $r^2 = 4 \Rightarrow r = 2$ (As $r \neq -2$ because x_1, x_2, x_3, x_4 are in increasing G.P.)

For $r = 2, a = 1$

So, $\alpha = a^2 r = (1)^2 \cdot 2 = 2$ and $\beta = a^2 r^5 = (1)^2 (2)^5 = 32$

Hence, $(\alpha + \beta) = 2 + 32 = 34$.

Now verify alternatives.

Q.21 (A) Q; (B) R; (C) S; (D) P

(A) $1 - 8 (\log_{10} x)^2 = \log_{10} x - 2 (\log_{10} x)^2 \Rightarrow 6 (\log_{10} x)^2 + \log_{10} x - 1 = 0$

$$\log_{10} x = \frac{1}{3}, -\frac{1}{2}$$

$$x = 10^{1/3}, 10^{-1/2}$$

$$\alpha = 10^{-1/2}, \beta = 10^{1/3}$$

$$\frac{(\beta^3 + 1)}{\alpha^2 \beta^3} = \frac{(10 + 1)}{10^{-1} 10} = 11$$

(B) For any $x > 0$ L.H.S. > 0 and R.H.S. < 0

No value of x will satisfy the given equation. Hence no solution.

(C) $|x| + \left| \frac{4 - x^2}{x} \right| = \left| x + \frac{4 - x^2}{x} \right|$

$$\text{as } |a| + |b| = |a + b| \Rightarrow ab \geq 0$$

$$x \left(\frac{4 - x^2}{x} \right) \geq 0, x \neq 0$$

$$\Rightarrow x^2 - 4 \leq 0 \Rightarrow x \in [-2, 2] - \{0\}$$

(D) $\log_b 125 = \frac{\log_e 8}{\log_e 2} \Rightarrow \log_b 125 = 3$

$$\Rightarrow b^3 = 125 \Rightarrow b = 5 \text{ Ans.}$$

Q.22 [0005]

$$r + s + t = 0, \quad rst = \frac{-2008}{8} = -251$$

Now Let $r + s = x$

$$s + t = y$$

$$t + r = z$$

$$x + y + z = 2(r + s + t) = 0$$

$$x^3 + y^3 + z^3 = 3xyz$$

$$= 3(r + s)(s + t)(t + r) = 3(-t)(-r)(-s) = -3rst = -3 \times (-251) = 753 \text{ Ans.}$$

Q.23 [0006]

Given that 9 is a root of the equation

$$\log_{\pi}(x^2 + 15a^2) - \log_{\pi}(a - 2) = \log_{\pi} \frac{8ax}{a - 2}, \quad a > 2$$

Putting $x = 9$,

$$\log_{\pi} \frac{81 + 15a^2}{a - 2} = \log_{\pi} \frac{72a}{a - 2} \Rightarrow 5a^2 - 24a + 27 = 0$$



i.e., $a = 3$ or $\frac{9}{5}$

$\therefore \log_{\pi}(x^2 + 135) = \log_{\pi}(24x)$

i.e., $x^2 - 24x + 135 = 0$.

Roots are 9 and 15.

The other root is 15. **Ans.**

Q.24 [0014]

Let $P(x) = ax^3 + bx^2 + cx + d$

$$P\left(\frac{1}{2}\right) + P\left(-\frac{1}{2}\right) = 2b\left(\frac{1}{4}\right) + 2d$$

$P(0) = d$

$$\frac{2b/4 + 2d}{d} = 100 \Rightarrow \frac{2b}{4d} + 2 = 100 \Rightarrow \frac{b}{2d} = 98$$

Also $\alpha + \beta + \gamma = -\frac{b}{a}$, $\alpha\beta\gamma = -\frac{d}{a}$ **Ans.**

Q.25 [0012]

$6 \leq a < 6.25$ $[a] = 6$

$\Rightarrow 2, h_1, \dots, h_{20}, 6 \rightarrow \text{H.P.}$

$h_{18} = \frac{14}{3}$

$2, a_1, \dots, a_{20}, 6 \rightarrow \text{A.P.}$

$a_3 = \frac{18}{7}$

$a_3 h_{18} = 12$ **Ans.**

